

Steps

22. $Al(Veneesha)$
 23. $DM(Keeshaun) \Rightarrow Al$
 (Keeshaun)

Reason

Modus ponens on 8, 21
 Universal instantiation on 2

24. $Al(Keeshaun)$

Modus ponens on 5, 23

25. $Al(Melissa) \wedge Al(Aaron)$

Conjunction on 16, 18

26. $Al(Melissa) \wedge Al(Aaron)$

Conjunction on 25, 20

$\wedge Al(Ralph)$

Conjunction on 26, 22

27. $Al(Melissa) \wedge Al(Aaron)$
 $\wedge Al(Ralph) \wedge Al(Veneesha)$

Conjunction on 27, 24

28. $Al(Melissa) \wedge Al(Aaron)$
 $\wedge Al(Ralph) \wedge Al(Veneesha)$
 $\wedge Al(Keeshaun)$

c) All movies produced by John Sayles are wonderful. John Sayles produced a movie about coal miners. Therefore, there is a wonderful movie about coal miners.

Soln:

$JS(x)$: x is a movie produced by John Sayles
 $W(x)$: Movie x is wonderful
 $CM(x)$: x is a movie about coal miners

Domain: The set of all movies.

Premise: 1. $\forall x(JS(x) \Rightarrow W(x))$ 2. $\exists x(JS(x) \wedge CM(x))$

Conclusion: $\exists x(W(x) \wedge CM(x))$

StepsReason

1. $\forall x(JS(x) \Rightarrow W(x))$

Premise

2. $\exists x(JS(x) \wedge CM(x))$

Premise

3. $JS(c) \wedge CM(c)$ for some
 c in the domain

Existential instantiation on 2